

ECO 362: Financial Investments

Final Review

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Capital asset pricing model

- The expected return on asset i is

$$\mathbb{E}[R_i - R_b] = \beta_i \mathbb{E}[R_m - R_b]$$

where

- R_b is the riskless rate.
- R_m is the market return.
- $\beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}$ is asset i 's market beta.
- An investor holding a risky asset (i.e., high beta) must be compensated through a higher expected return (i.e., low price).

Present-value formula

- An asset costs P_i today and pays a random cash flow X_i at a future date (e.g., in 10 years).
- According to the CAPM,

$$P_i = \frac{\mathbb{E}[X_i]}{1 + R_b + \beta_i \mathbb{E}[R_m - R_b]}$$

- $R_b + \beta_i \mathbb{E}[R_m - R_b]$ is a risk-adjusted interest rate, which is increasing in β_i .

Gordon growth model

- Price-dividend ratio is

$$\frac{P_0}{D_0} = \frac{1 + G}{\bar{R} - G}$$

- $\bar{R}$ is expected return (per year).
 - G is expected dividend growth (per year).
- When the price-dividend ratio is high, either
 1. Expected returns are low: Forecasts low future returns.
 2. Expected dividend growth is high: Forecasts high future dividend growth.

Multi-factor model

- Expected return on asset i is

$$\mathbb{E}[R_i - R_b] = \sum_{j=1}^J \beta_{i,j} \mathbb{E}[F_j - R_b]$$

- $\beta_{i,j}$ is asset i 's beta with respect to risk factor j .
 - $\mathbb{E}[F_j - R_b]$ is expected excess return on risk factor j .
- CAPM:

$$\mathbb{E}[R_i - R_b] = \beta_{i,m} \mathbb{E}[R_m - R_b]$$

- Fama-French 3-factor model:

$$\mathbb{E}[R_i - R_b] = \beta_{i,m} \mathbb{E}[R_m - R_b] + \beta_{i,SMB} \mathbb{E}[SMB] + \beta_{i,HML} \mathbb{E}[HML]$$

Optimal portfolio choice with labor income

- Allocate financial wealth A between stocks S and riskless asset $A - S$.
- Human capital H is the present value of future labor income.
- Stocks have mean return μ and variance σ^2 .
- Riskless interest rate is R_b .
- Optimal portfolio choice is

$$\frac{S}{A} = \left(1 + \frac{H}{A}\right) \frac{\mu - R_b}{\gamma \sigma^2}$$

Expectations hypothesis

- Continuously compounded t -year zero-coupon yield

$$y_0(t) = \underbrace{\frac{1}{t} \sum_{s=0}^{t-1} \mathbb{E}_0[y_s(1)]}_{\text{expected short rate}} + \underbrace{\frac{t-1}{t} \mu}_{\text{term premium}}$$

- If the short rate follows a mean-reverting process,

$$y_0(t) - \bar{y} = \underbrace{\frac{1}{t} \left(\frac{1 - \rho^t}{1 - \rho} \right)}_{b(t)} (y_0(1) - \bar{y}) + \underbrace{\frac{t-1}{t} \mu}_{a(t)}$$

- $\bar{y}$ is the long-run mean of the short rate.
- ρ determines the speed of mean reversion.

Break-even inflation

- Break-even inflation is the yield spread between nominal Treasury and TIPS.

$$\underbrace{y_b(T) - y_{TIPS}(T)}_{\text{break-even inflation}} = \underbrace{\frac{1}{T} \ln(\mathbb{E}[\exp(Ti_T)])}_{\text{expected inflation}} + \text{IRP}$$

- Inflation risk premium (IRP).
 - Zero if inflation risk is idiosyncratic.
 - If inflation is positively correlated with the SDF, nominal Treasuries are risky and require an extra risk premium.

Credit spread

- Credit spread is the yield spread between corporate bond and Treasury.

$$\underbrace{y_{CB}(T) - y_b(T)}_{\text{credit spread}} = \underbrace{-\frac{1}{T} \ln(\mathbb{E}[D_T])}_{\text{expected default}} + \text{CRP}$$

- Credit risk premium (CRP).
 - Zero if default risk is idiosyncratic.
 - If default is positively correlated with the SDF, corporate bonds are risky and require an extra risk premium.

Futures

- A contract to buy/sell an asset at a predetermined price F in T years from now.
 - F is the futures price.
 - T is maturity.
 - r_b is continuously compounded interest rate.
 - d is continuously compounded dividend rate.
- If the spot price is S_T at maturity,
 - Buyer's (long) payoff is $S_T - F$.
 - Seller's (short) payoff is $F - S_T$.
- By no arbitrage,

$$F = S_0 \exp((r_b - d)T)$$

Expected futures return

- Collateralized long futures strategy.
 1. Long futures at futures price F and maturity T .
 2. Lend $F \exp(-r_b T)$ for T years.
- Expected excess return is

$$\mathbb{E}[r_p - r_b] = \underbrace{\frac{1}{T} \mathbb{E} \left[\ln \left(\frac{S_T}{S_0} \right) \right]}_{\text{expected spot return}} + \underbrace{\frac{1}{T} \ln \left(\frac{S_0}{F} \right)}_{\text{basis}}$$

- Backwardation if $S_0 > F$: Earn a higher return because futures is priced below the spot.
- Contango if $S_0 < F$: Earn a lower return because futures is priced above the spot.

Interest rate swap

- A contract to exchange a floating rate $R_{b,t}$ for a fixed rate c for T years.
 - $R_{b,t}$ is a short-term interest rate.
 - T is maturity.
- Let N be the notional amount.
- On each payment date $t = 1, \dots, T$,
 - Counterparty that receives floating and pays fixed: $(R_{b,t} - c)N$.
 - Counterparty that receives fixed and pays floating: $(c - R_{b,t})N$.
- By no arbitrage, fixed rate c is the par yield at maturity T .

Call option

- Buyer pays price C for the right to buy an asset at exercise price X on expiration date.
- If the asset price is S_T on expiration date, payoff is
 - Buyer: $\max(S_T - X, 0)$
 - Seller: $-\max(S_T - X, 0)$

Put option

- Buyer pays price P for the right to sell an asset at exercise price X on expiration date.
- If the asset price is S_T on expiration date, payoff is
 - Buyer: $\max(X - S_T, 0)$
 - Seller: $-\max(X - S_T, 0)$

Black-Scholes model

- Call price is

$$C = S_0 \times \mathbb{N}(d_1) - \exp(-r_b T)X \times \mathbb{N}(d_2)$$

where

$$d_1 = \frac{\ln(S_0/X) + (r_b + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

- $\mathbb{N}(d)$ is probability that a standard normal random variable is less than d .
- By the put-call parity,

$$\begin{aligned} P &= C - S_0 + \exp(-r_b T)X \\ &= -S_0(1 - \mathbb{N}(d_1)) + \exp(-r_b T)X(1 - \mathbb{N}(d_2)) \end{aligned}$$

Cumulative return with taxes

- Taxable account.

$$(1 - \tau_{i,0}) \prod_{t=1}^T (1 + (1 - \tau_{i,t})R_t)$$

- Traditional 401(k).

$$(1 - \tau_{i,T}) \prod_{t=1}^T (1 + R_t)$$

- Roth 401(k).

$$(1 - \tau_{i,0}) \prod_{t=1}^T (1 + R_t)$$

Strategy for preparing

- The exam is cumulative and will cover material from the entire course.
- The cheat sheet must be letter size (8.5×11 in) with writing on both sides.
- You should be able to apply the formulas to solve actual problems with numbers.
- All questions are solvable with a hand calculator.